INCH POUND

MIL-S-83194A(USAF)
14 February 1994
SUPERSEDING
MIL-S-83194(USAF)
6 December 1968

MILITARY SPECIFICATION

SOCKS, MEN'S, ANTI-EXPOSURE, RUBBER SRU-25/P

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers rubber socks designated SRU-25/P.

1.2 Classification. The socks shall be one type and in the following widths and sizes, as specified (see 6.2):

SIZE	WIDTH	SIZE	WIDTH
5-1/2	N (B)	6	R (D)
7-1/2	N (B)	8	R (D)
9-1/2	N (B)	10	R (D)
11-1/2	N (B)	12	R (D)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: The Technology & Industrial Services Division, SA-ALC/TILDD, 485 Quentin Roosevelt Rd., Bldg 171, Post C-12, Kelly AFB, TX 78241-6524 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8475

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SPECIFICATIONS

FEDERAL

TT-T-548	Toluene, Technical
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-B-676	Boxes, Setup

MILITARY

MIL-C-5646	Cloth, Airplane
MIL-C-19002	Cloth, Coated, and Strip, Coated Cloth - Polychloroprene on Nylon

STANDARDS

FEDERAL

FED-STD-191	Textile Test Methods
FED-STD-601	Rubber, Sampling and Testing

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-831	Test Reports, Preparation of

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Service, Detachment Office, 700 Robbins Avenue, Bldg 4D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

Drawing

Air Force

62H4256	Patterns, Rubber Men's Anti-Exposure Socks

(Copies of drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as specified by the contracting officer.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the ssues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

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ASTM D301 Standard Test Methods for Soluble Cellulose Nitrate

ASTM D482 Standard Test Methods for Ash from Petroleum Products

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103-1137.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Preproduction. When required (see 4.3 and 6.2), this specification makes provisions for preproduction testing.

3.2 Materials.

3.2.1 Rubber compound. The rubber compound for the socks shall be made from all new natural rubber (no reclaimed rubber shall be permitted). The physical properties of the compound shall conform to the requirements specified in Table I.

TABLE I. Physical Properties of Component Materials

Property	Requirements
Rubber compound	
Tensile strength, minimum	2,000 pounds per square inch (psi)
Hardness (durometer)	35 plus or minus 5
Color	black
Flocking cotton	
Color	white
Fiber lengths	0.021 - 0.029 inch
Ash content, maximum	0.2 percent
Nitrate content, maximum	0.2 percent

3.2.2 Flocking material. The flocking material shall be cotton and shall conform to the requirements specified in Table I.

3.2.3 Physical requirements. The physical requirements of finished socks shall conform to Table II.

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TABLE II. Physical Properties of Socks

Property	Requirements After Flocking
Thickness	0.020 (plus 0, minus 0.003) inch
Flocked sock	0.008 inch (maximum)
Flocking	0.005 inch (minimum)
Tension strength	3,000 psi (minimum)
Elongation	800 percent (minimum)
Hardness (durometer)	35 plus or minus 5
Flocking adhesion	5 pounds, 1 inch width (minimum)
Ultraviolet	no evidence of deterioration

3.3 Design and construction. The design of the dipped or molded socks shall conform to Figure 1, and the size dimensions for the molds used for the socks shall conform to Table III. The bottom shape of the socks and the toe and heel outlines shall conform to the patterns and templates referenced on Drawing 62H4256. Variations of plus or minus 1/16 inch in the width dimension and plus or minus 1/32 inch in the length dimension of the bottom will be permitted. The height of all sizes of socks shall be 9-1/4 inches (plus or minus 1/8 inch) when measured at the back on the inside. The inside top circumference of all sizes of the socks shall be the same (18 inches, plus or minus 1/16 inch). The ankle line and the heel taper line shall be located 3-1/2 inches (plus or minus 1/8 inch) from the bottom of the sock. The shape of the sock at the ankle line shall conform to Figure 1. The dimensions referenced on Figure 1 are for the size 7-1/2N. The grade for increase or decrease of the dimensions between the indicated points for larger or smaller sizes, as specified herein, shall be 3/64 inch. The socks shall be completely cotton flocked on the inside and the outside. Unless otherwise specified herein, the flocking on the inside and the outside of the socks shall be applied before the socks are cured and without the use of adhesives. At the option of the contractor, the flocking on the inside of the socks may be applied after the socks are cured and by the use of adhesives.

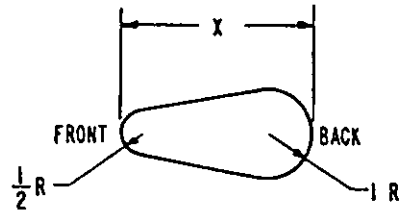
TABLE III. Size Dimensions of Molds

Size	Width	Ball Girth 1/	Waist Girth 1/	Instep Girth 1/	Heel Girth 1/	Toe Girth I/
5-1/2	N (B)	8-1/16	8	8-1/2	12-5/16	5-1/4
7-1/2	N (B)	8-9/16	8-1/2	9	13-1/16	5-3/4
9-1/2	N (B)	9-1/8	9-1/16	9-9/16	13-7/8	6-1/4
11-1/2	N (B)	9-3/4	9-11/16	10-3/16	14-3/4	6-3/4
6	R (D)	8-3/4	8-5/8	9-1/8	12-3/4	5-3/8
8	R (D)	9-1/4	9-3/16	9-11/16	13-1/2	5-7/8
10	R (D)	9-13/16	9-13/16	10-5/16	14-5/16	6-3/8
12	R (D)	10-7/16	10-1/2	11	15-1/4	6-7/8

1/ With the exception of the heel girth, all girth dimensions include the sole width.

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SECTION A-A

X DETERMINED FROM HEEL GIRTH DIMENSION AT THE FRONT, HEEL TEMPLATE AND THE ANKLE LINE FOR EACH SIZE

UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES

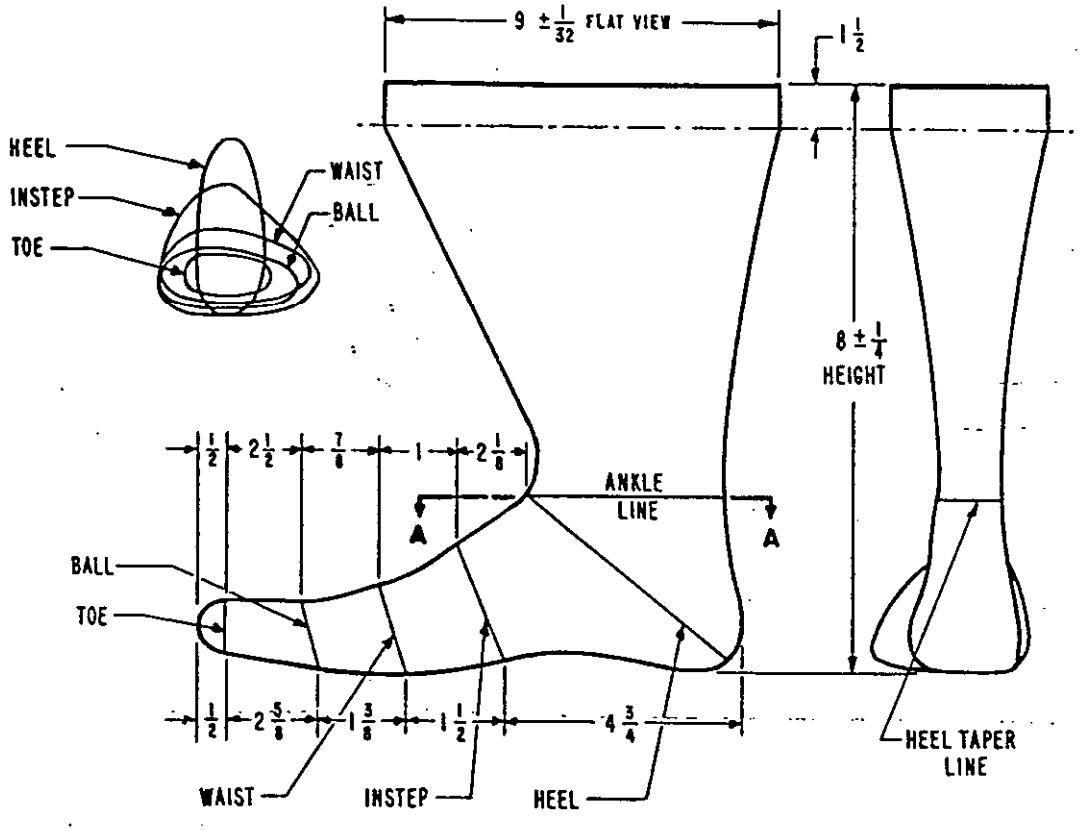


FIGURE 1. Socks

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3.4 Performance.

3.4.1 Leakage. The socks shall not leak when immersed in water within 1 inch (plus or minus 1/8 inch) from the top edge.

3.5 Identification of product. Each sock shall be permanently and legibly marked in accordance with MIL-STD-130. The marking shall be on the inside at the top, 1 inch (plus or minus 1/8 inch) from the top edge, and shall include the following specific information:

SOCKS, MEN'S ANTI-EXPOSURE, RUBBER SRU-25/P
Size 1/_____
SPECIFICATION MIL-S-83194 (USAF)

1/ The manufacturer shall insert the applicable size.

3.6 Workmanship. Inasmuch as the correct functioning of these socks directly affects the safety of using personnel, workmanship shall be equal to or better than the highest grade of manufacturing practices to ensure minimum opportunity for the development of defects while in service.

4. QUALITY ASSURANCE

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of tests. The inspection and testing of socks shall be classified as follows:

a. Preproduction tests

b. Quality conformance tests.

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4.3 Preproduction tests. The preproduction tests shall consist of all of the inspections and tests described under 4.5.

4.3.1 Preproduction test sample. The preproduction sample, which shall consist of a minimum of three pairs of socks of different sizes and widths, shall be subjected to the preproduction tests. When the contractor is in continuous production of items covered by this specification from contract to contract or within a 12-month period, submission of preproduction samples on the new contract may be waived at the discretion of the procuring activity (see 6.2). Approval of the preproduction test samples or waiver of the preproduction test samples shall not preclude the requirement for quality conformance tests.

4.3.2 Preproduction test report. When specified on DD Form 1423 and by the procuring activity (see 6.2), three copies of the preproduction test report in the format specified in MIL-STD-831 shall be submitted with the preproduction samples specified in 4.3.1.

4.4 Quality conformance tests. Quality conformance tests shall consist of:

a. Individual tests

b. Sampling tests.

4.4.1 Individual tests. Each sock shall be subjected to the following tests as described under 4.5:

a. Examination of product

b. Leakage.

4.4.2 Sampling tests. One sock from every two days of production shall be subjected to the following tests as described under 4.5:

a. Thickness (flocked sock) (see Table IV)

b. Tensile strength

c. Hardness (durometer)

d. Flocking adhesion.

TABLE IV. Physical and Chemical Properties of Material

Property	Test Method
Thickness (flocked sock)	5030 of FED-STD-191
Hardness (durometer	3021 of FED-STD-601
Tensile strength at break	4111 of FED-STD-601
Elongation	4121 of FED-STD-601
Tension set	4411 of FED-STD-601
Ash content	ASTM Standard D 482
Nitrate content	ASTM Standard D 301
Ultraviolet	(see 4.5.4)
Flocking adhesion	(see 4.5.5)

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4.4.2.1 Rejection and retest. When an item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the tests shall be repeated.

4.4.2.2 Individual tests may continue. For production reasons, individual tests or other sampling plans may be continued pending the investigation of sampling test failure. Final acceptance of items on hand or produced later shall not be made until it is determined that all items meet all the requirements of the specification.

4.4.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.5 Test methods.

4.5.1 Examination of product. Each sock shall be examined to determine compliance with the requirements of this specification with respect to material, workmanship, and marking.

4.5.2 Leakage. Each sock shall be tested individually and shall not show evidence of leaks when immersed in water to within 1 inch (plus or minus 1/8 inch) from the top edge.

4.5.3 Materials. The physical and chemical properties of the material shall be tested and shall meet the requirements specified in Section 3. The methods of testing shall be as specified in Table IV.

4.5.4 Ultraviolet. A specimen of the sock shall be subjected to ultraviolet light for a period of 4 hours. At the end of this period there shall be no evidence of deterioration.

4.5.5 Flocking adhesion.

4.5.5.1 Test specimen. The materials required for the preparation of the adhesion specimen shall be adhesion conforming to International Corp. No. 533-1 Parts A and B or Union Bay State Chemical Co. No. N-524 Parts A and B or their equivalent; bias tape, 1-3/16 inches wide, conforming to type III of MIL-C-19002; and tape backing conforming to MIL-C-5646 or any other suitable, lightweight, untreated, cotton cloth. The adhesion test specimen shall be prepared as follows: Cut strips of sufficient width and 6 inches long from the socks. Clean the flock surface by washing three times with a cloth saturated with toluene conforming to TT-T-548. Use a clean cloth for each washing. Allow to dry for 15 minutes before applying adhesive. Apply three coats of adhesive to the cleaned flock surface. Drying time between coats and before bonding shall be in accordance with the adhesive manufacturer's instructions. Remove the holland cloth backing from the previously prepared tape laminate. Apply tape to the cemented surface and roll firmly. Condition the cemented seams in accordance with instructions provided by the adhesive manufacturer.

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4.5.5.2 Adhesion test. After conditioning, the specimen shall be cut to 1 inch width, and strip adhesion shall be determined in accordance with the procedure for strip specimen described in method 8011 of FED-STD-601. The results shall be the average of the three specimens.

4.5.6 Thickness (flocking). The thickness of each sock shall be determined in accordance with method 2011 of FED-STD-601. The flocking on the inside and the outside of each sock shall be removed, by the use of a solvent or by buffing, over a 1-square-inch area at each of six (minimum) points: four on the upper of the sock, equally spaced around the upper, and two on the sole area. The average thickness of the unflocked rubber of the sock at these points shall be determined in accordance with method 2011 of FED-STD-601. The average thickness of the unflocked rubber of the sock shall be subtracted from the average thickness of the finished socks, and the results shall be recorded as the average thickness of the flocking.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. If deemed necessary by the manufacturer, any creases in the socks shall be dusted with a commercial talc of the type used for rubberized materials. In addition, wrapping tissue shall be placed in the folds, and the socks shall be overwrapped with tissue so that the tissue in the folds will be maintained in place. A matched pair of individually wrapped socks shall then be placed in a snug-fitting box conforming to PPP-B-676. The closure of the container shall be in accordance with the appendix of the container specification.

5.1.2 Level C. Socks shall be preserved and packaged to afford adequate protection, against deterioration and physical damage, during shipment from the supply source to the first receiving activity for immediate use. This level may conform to the supplier's commercial practice provided the latter meets the requirements.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Twenty-four pairs of socks of one size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container conforming to class - weather-resistant, style RSC OF PPP-B-636. The containers shall be closed and reinforced in accordance with the appendix of the container specification. If the quantity of socks of the same size is less than the quantity of socks required per container, mixed sizes (see 5.3.1) may be packed within the same shipping container.

5.2.2 Level B. Level B packing shall be the same as specified in 5.2.1, except that the shipping container shall conform to class - domestic.

5.2.3 Level C. Socks shall be packed to afford adequate protection against damage or deterioration, at the lowest rate, during direct shipment from the supply source to the first receiving activity for immediate use. The containers shall conform to the applicable rules of carriers and the regulations applicable to the mode of transportation.

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5.3 Marking. In addition to any special markings required by the contract or order, interior and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Mixed sizes. Each shipping container that is packed with mixed sizes (see 5.2.1) shall have securely attached to the end and side, directly under the printing or stenciling, a white paper label 5 by 4 inches with the words mixed sizes plainly stamped or printed thereon. Under these words shall be legibly stamped or printed the quantities and sizes of socks contained therein.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The socks covered by this specification are intended for use with the coveralls, flying, anti-exposure CWU-21/P; the underwear, double-knitted, anti-exposure CWU-22/ P; and the liner, anti-exposure, coveralls, flying, CWU-23/P. The socks are intended to be a waterproof sealing-medium for the leg endings of the coveralls to complete the anti-exposure protection where water temperatures are expected to be below plus 50 degrees Fahrenheit.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification

b. Size required (see 1.2 and 3.3)

c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2)

d. Waiver of preproduction test samples, if applicable (see 3.1 and 4.3.1)

e. Preproduction test report requirement (see 4.3.2)

f. Selection of applicable levels of preservation and packaging; and packing (see 5.1 and 5.2).

6.3 Subject term (key word) listing.

Rubber compound
Flocking material
Color
Fiber Lengths
Thickness
Tension strength
Toe
Heel
Ankle length
Ash content
Nitrate content
Leakage
Adhesion

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6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force - 99

Preparing activity:
Air Force - 82

(Project 8475-F172)

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